

A quasimorphism-metric counterexample to the Lip_0 almost-invariant approximation question

agent_lane_02

2026-07-18

Status. Candidate counterexample, likely valid.

Source paper. Tomas Raunig, *Almost-invariant elements of group actions on Lip_0 spaces*, arXiv:2506.06770, v1, submitted 7 June 2025 [1].

1 Source question

The target is Question 1.3 of [1], on page 2. It asks whether, for an isometric action of a discrete group G on a pointed metric space $(M, d, 0)$, every norm-one $f \in \text{Lip}_0(M)$ that is $\delta/10$ -invariant for the induced action

$$\alpha(g, f)(x) = f(g^{-1}x) - f(g^{-1}0)$$

must be within distance δ of an invariant element of $\text{Lip}_0(M)$.

Using this fact, we arrive to an equivalent formulation of Question 1.1 in the context described above:

Question 1.3. Let $\delta > 0$, $(M, d, 0)$ be a pointed metric space and G be a discrete group acting on M by isometries. Consider the action of G by linear isometries $\alpha : G \times \text{Lip}_0(M) \rightarrow \text{Lip}_0(M)$ given by

$$\alpha(g, f)(x) = f(g^{-1}x) - f(g^{-1}0), \quad g \in G, f \in \text{Lip}_0(M), x \in M.$$

Assume $f \in \text{Lip}_0(M)$ with $L(f) = 1$ is $\delta/10$ -invariant with respect to the action α , that is $\|\alpha(g, f) - f\| = \|x \mapsto f(g^{-1}x) - f(x)\| \leq \delta/10$ for every $g \in G$. Must there exist invariant $\bar{f} \in \text{Lip}_0(M)$ with $\|f - \bar{f}\| \leq \delta$?

Here it is appropriate to acknowledge that not every isometric action on a Lipschitz-free

2 Proof Intuition

For a group acting on itself by left translations, Raunig proves that the invariant elements of $\text{Lip}_0(G)$ are exactly the real homomorphisms $G \rightarrow \mathbb{R}$. So we choose a group with no nonzero real homomorphisms but with a nonzero homogeneous quasimorphism q .

The defect of q says that

$$q(a^{-1}x) - q(a^{-1}) - q(x)$$

is uniformly bounded in x . If we put a large discrete term into a left-invariant metric, every such bounded defect function has very small Lipschitz constant. At the same time, including the term $|q(x^{-1}y)|$ in the metric keeps q itself at Lipschitz norm one. Thus q is almost invariant but cannot be close to any invariant function, because the invariant functions are all zero.

3 Counterexample

Theorem 1. *Question 1.3 of [1] has a negative answer. More precisely, there are a discrete group G , a pointed metric space $(M, d, 0)$ on which G acts by isometries, a number $\delta_0 = 1/2$, and an element $f \in \text{Lip}_0(M)$ with $L(f) = 1$ such that*

$$\sup_{a \in G} \|\alpha(a, f) - f\| \leq \frac{\delta_0}{10},$$

but no invariant $\bar{f} \in \text{Lip}_0(M)$ satisfies $\|f - \bar{f}\| \leq \delta_0$.

Proof. Let $G = \text{PSL}(2, \mathbb{Z})$. This group is isomorphic to $C_2 * C_3$, so its abelianization is finite; hence

$$\text{Hom}(G, \mathbb{R}) = 0.$$

It is also a non-elementary hyperbolic group. By the standard Epstein–Fujiwara/Bestvina–Fujiwara quasimorphism theorem, in the form recorded for non-elementary hyperbolic groups in [3, Theorem B], G admits a nonzero homogeneous quasimorphism $q : G \rightarrow \mathbb{R}$. Put

$$D = \sup_{u, v \in G} |q(uv) - q(u) - q(v)|.$$

Since G has no nonzero real homomorphism, this q is not a homomorphism, and therefore $D > 0$.

Set $B = 100D$. Define $\rho : G \rightarrow [0, \infty)$ by

$$\rho(e) = 0, \quad \rho(t) = B + |q(t)| + D \quad (t \neq e),$$

and define $d(x, y) = \rho(x^{-1}y)$. We first check that this is a left-invariant metric. Positivity and left invariance are immediate. Since q is homogeneous, $q(t^{-1}) = -q(t)$, so $\rho(t^{-1}) = \rho(t)$ and d is symmetric. For the triangle inequality, only the case $a, b, ab \neq e$ needs attention:

$$\rho(ab) = B + |q(ab)| + D \leq B + |q(a)| + |q(b)| + 2D \leq \rho(a) + \rho(b).$$

All cases where one of a, b, ab is the identity are trivial.

Now take the pointed metric space $M = (G, d, e)$, with G acting on M by left translations. This action is by isometries because d is left invariant. We claim that $q \in \text{Lip}_0(M)$ and $L(q) = 1$. If $x \neq y$ and $a = x^{-1}y$, then the quasimorphism defect gives

$$|q(y) - q(x)| = |q(xa) - q(x)| \leq |q(a)| + D \leq \rho(a) = d(x, y),$$

so $L(q) \leq 1$. Since q is nonzero, choose $g \in G$ with $q(g) \neq 0$. Homogeneity gives

$$\frac{|q(g^n) - q(e)|}{d(e, g^n)} = \frac{n|q(g)|}{B + n|q(g)| + D} \rightarrow 1,$$

and hence $L(q) = 1$.

For $a \in G$, let

$$\psi_a(x) = \alpha(a, q)(x) - q(x) = q(a^{-1}x) - q(a^{-1}) - q(x).$$

By the definition of D , $|\psi_a(x)| \leq D$ for all x . Thus, for $x \neq y$,

$$|\psi_a(x) - \psi_a(y)| \leq 2D \leq \frac{2D}{B + D} d(x, y) = \frac{2}{101} d(x, y).$$

Therefore

$$\|\alpha(a, q) - q\| \leq \frac{2}{101} < \frac{1}{20} = \frac{\delta_0}{10}$$

for every $a \in G$.

Finally, Raunig’s Corollary 2.3 says that for left translations on a group with a left-invariant metric, the invariant elements of $\text{Lip}_0(G)$ are exactly the homomorphisms $G \rightarrow \mathbb{R}$ [1, Corollary 2.3]. Since $\text{Hom}(G, \mathbb{R}) = 0$, the only invariant element is 0. But

$$\|q - 0\| = L(q) = 1 > \frac{1}{2} = \delta_0.$$

Thus the conclusion of Question 1.3 fails. □

4 Verification Notes

No computational search is used. The proof is symbolic. The numerical constant check is

$$\frac{2D}{100D + D} = \frac{2}{101} < \frac{1}{20}.$$

The construction does not contradict the positive cases in [1]: the metric is not a word metric, and the group is not amenable. The result only refutes the arbitrary pointed-metric-space formulation of Question 1.3.

5 Novelty Check

The local run indexes were searched on 2026-07-18 for the exact arXiv id ‘2506.06770’ and for the phrases “almost invariant”, “Lip0”, “Lipschitz-free”, “Kazhdan”, “Yom Din”, “quasimorphism”, “amenability”, and related fixed-point terminology. No exact packet or close duplicate of this construction was found. Nearby entries concern the original Glasner question, fixed point properties, invariant means, amenability, property (T), and other Lipschitz-free questions.

External web search on 2026-07-18 checked the exact title and arXiv id, “Question 1.3 Lip0 invariant quasimorphism”, “Lip0 almost invariant quasimorphism”, and counterexample variants. I found the source preprint [1], the Glasner background paper [2], and general homogeneous-quasimorphism literature [3], but no later work explicitly answering the Lip0 question.

6 Human Review Recommendation

Review as a likely valid counterexample. The main points to check are: the left-invariant metric triangle inequality, the defect estimate for $\alpha(a, q) - q$, and the cited existence of a nonzero homogeneous quasimorphism on a group with no real characters.

References

- [1] T. Raunig, *Almost-invariant elements of group actions on Lip_0 spaces*, arXiv:2506.06770, 2025.
- [2] E. Glasner, *On a question of Kazhdan and Yom Din: With an appendix by Nicolas Monod*, Israel Journal of Mathematics 251 (2022), 467–493.
- [3] F. Fournier-Facio and R. D. Wade, *Aut-invariant quasimorphisms on groups*, arXiv:2211.00800, to appear in Transactions of the American Mathematical Society.